

# CS5340 - Uncertainty Modelling in AI

## Lecture 1: Introduction to Probabilistic Reasoning

Scribed by: Claude (Anthropic) | Curated by: Assoc. Prof. Lee Gim Hee  
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## 1 Lecture Overview

- **Theme:** This lecture builds the mathematical foundations needed to handle **uncertainty** in AI systems using **probabilistic reasoning**.
- **Why it matters:** These tools underpin later topics in the course — Bayesian networks, Markov random fields, hidden Markov models, sampling methods, and variational inference.
- **Roadmap of this lecture:**
  1. Probability spaces, random variables, and probability distributions.
  2. The rules of probability: joint, marginal (sum rule), conditional (product rule), Bayes' rule, independence, and expectation.
  3. Common probability distributions: Bernoulli, categorical, univariate/multivariate normal.
  4. Conjugate distributions and why they matter for Bayesian parameter learning.

### Learning Outcomes

By the end of this lecture, you should be able to:

1. Describe uncertain quantities using random variables and joint probabilities.
2. Explain the basic rules of probability — sum, product, Bayes', independence, and expectation rules.
3. Use common probability distributions — Bernoulli, categorical, univariate and multivariate normal.
4. Explain the use of conjugate distributions.

## 2 Probability Space

### Definition : Probability Space

A **probability space** is a triple  $(\Omega, E, P)$  that models a process consisting of outcomes that occur **randomly**:

- **Outcome/sample space  $\Omega$ :** the agreed-upon set of all possible outcomes.
- **Event space  $E \subseteq 2^\Omega$ :** a subset of the power set of  $\Omega$ , i.e. the set of *measurable events* to which we assign probabilities.
- **Probability distribution  $P : E \rightarrow \mathbb{R}$ :** assigns a probability value to each event.

## 2.1 Properties of the Event Space

The event space  $E$  must satisfy three basic properties:

1. It must contain the **empty event**  $\emptyset$  and the **trivial event**  $\Omega$ .
2. It is **closed under countable unions**: if  $\alpha_i \in E$  for  $i = 1, 2, \dots$ , then  $\bigcup_{i=1}^{\infty} \alpha_i \in E$ .
3. It is **closed under complements**: if  $\alpha \in E$ , then  $\Omega - \alpha \in E$ .

### Worked Example: Event Spaces

**Dice roll.**  $\Omega = \{1, \dots, 6\}$ . A valid event space is  $E = \{\{1, 3, 5\}, \{2, 4, 6\}, \emptyset, \Omega\}$  (odd vs. even outcomes). One can check it contains  $\emptyset$  and  $\Omega$ , is closed under unions ( $\{1, 3, 5\} \cup \{2, 4, 6\} = \Omega \in E$ ), and closed under complements ( $\Omega - \{2, 4, 6\} = \{1, 3, 5\} \in E$ ).

**Lightbulb lifetime.**  $\Omega = [0, \infty)$ . A valid event space is  $E = \{[0, 90), [90, \infty), \emptyset, \Omega\}$ , e.g. the event “lifespan  $\geq 90$ ”.

**Marble draws.** Picking 2 marbles (one at a time) from a bag of blue/red marbles gives sample space  $\Omega = \{BB, BR, RB, RR\}$ , easily visualized with a tree diagram or a  $2 \times 2$  table.

### Key Rule: Axioms of Probability

A probability distribution  $P$  over  $(\Omega, E)$  is a mapping  $P : E \rightarrow \mathbb{R}$  satisfying:

1. **Non-negativity:**  $P(\alpha) \geq 0$  for all  $\alpha \in E$ .
2. **Normalization:**  $P(\Omega) = 1$ .
3. **Additivity for disjoint events:** if  $\alpha, \beta \in E$  and  $\alpha \cap \beta = \emptyset$ , then  $P(\alpha \cup \beta) = P(\alpha) + P(\beta)$ .

## 3 Random Variables

### Definition : Random Variable

A **random variable**  $X$  (upper case) is the formal machinery for discussing attributes and their values across outcomes. Formally, given a probability space  $(\Omega, E, P)$ , a random variable is a function

$$X : \Omega \rightarrow S$$

that maps a set of possible outcomes  $\Omega$  to a measurable space  $S$  (typically  $S \in \mathbb{R}$ ).

- $Val(X)$  denotes the **set of values** that  $X$  can take.
- A lower-case letter  $x$  denotes a generic **realization** (specific value) of  $X$ ; we write  $P(X = x)$  for all  $x \in Val(X)$ , often shortened to  $P(x)$ . The notation  $x^i$  denotes a specific value of  $X$ .
- Random variables can be:
  - **Discrete** — takes values from a predefined (finite/countable) set.  
*Examples:* rolling a die,  $Val(X) = \{1, \dots, 6\}$ ; weather condition,  $Val(X) = \{\text{rain, cloud, snow, sun, wind}\}$ ; number of people on a train,  $Val(X) = \mathbb{Z}_{\geq 0}$ .
  - **Continuous** — takes values that are real numbers.  
*Examples:* time to finish an exam,  $Val(X) = [1, 2]$  hours; height of a tree,  $Val(X) = \mathbb{R}_{>0}$ ; ambient temperature,  $Val(X) = \mathbb{R}$ .

## 4 Probability Distributions: Discrete vs. Continuous

**Discrete: Probability Mass Function (PMF)**

Denoted  $P(x)$ , defined so that

$$\sum_{i=1}^K P(X = x^i) = 1, \quad 0 \leq P(X = x^i) \leq 1 \quad \forall i = 1, \dots, K, \quad K = |\text{Val}(X)|.$$

**Continuous: Probability Density Function (PDF)**

Denoted (lower case)  $p(x) : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$ , satisfying  $\int_{\text{Val}(X)} p(x) dx = 1$ . The corresponding **cumulative function** is

$$P(X \leq a) = \int_{-\infty}^a p(x) dx, \quad P(a \leq X \leq b) = \int_a^b p(x) dx, \quad P(X = x^i) = \int_{x^i}^{x^i} p(x) dx = 0.$$

**Notation Abuse**

In this course, we abuse notation by denoting **both** the PMF and the PDF with the lower-case  $p(x)$ ! We silently note the property differences in  $P(x)$  depending on whether  $X$  is discrete or continuous.

## 5 Probabilistic Reasoning: Modeling and Inference

**Definition : Probabilistic Modeling**

The central paradigm of probabilistic reasoning is to:

1. Identify all relevant variables  $X_1, \dots, X_N$  of the environment, and
2. make a probabilistic model  $p(X_1, \dots, X_N)$  of their interactions.

**Definition : Probabilistic Inference**

Reasoning (**inference**) is then performed by:

1. introducing **evidence** that sets variables to a known state, and
2. subsequently computing probabilities of interest, **conditioned on** this evidence.

To do this, we need the definitions of **joint probability**, **marginalization**, **conditional probability**, **Bayes' rule**, and **independence** — covered next. This lecture applies these to a small number of variables; later lectures extend them to **graphical models** over many variables.

## 6 Joint Probability

**Definition : Joint Probability**

Consider all combinations of events of two random variables  $X$  and  $Y$  — some combinations of outcomes are more likely than others. This is captured by the **joint probability distribution**  $p(x, y)$ , read as “probability of  $X$  and  $Y$ ”. It generalizes to more than two variables:  $p(a, b, c, \dots)$ . Joint distributions can be discrete, continuous, or a discrete-continuous mixture.

## 7 Marginalization (Sum Rule)

**Key Rule: Marginalization / Sum Rule**

Recover the probability distribution of any variable in a joint distribution by **integrating (or summing) over all other variables**:

$$\textbf{Continuous: } p(x) = \int p(x, y) dy, \quad p(y) = \int p(x, y) dx$$

$$\textbf{Discrete: } p(x) = \sum_y p(x, y), \quad p(y) = \sum_x p(x, y)$$

$$\textbf{Discrete-continuous: } p(x) = \sum_y p(x, y), \quad p(y) = \int p(x, y) dx$$

This works in higher dimensions too, e.g.  $p(x, y) = \sum_w \int p(w, x, y, z) dz$ .

## 8 Conditional Probability (Product Rule)

**Definition : Conditional Probability**

$p(x | Y = y^*)$  reads “probability of  $X$  given  $Y = y^*$ ” — the **relative propensity** of  $X$  to take different outcomes given that  $Y$  is fixed to value  $y^*$ . It is extracted from the joint probability by taking the appropriate slice and **normalizing** (so the area/sum is 1):

$$P(x | Y = y^*) = \frac{p(x, Y = y^*)}{\int p(x, Y = y^*) dx} = \frac{p(x, Y = y^*)}{p(Y = y^*)}.$$

**Key Rule: Product Rule / Chain Rule**

Compactly,  $p(x | y) = \frac{p(x, y)}{p(y)}$ , which rearranges to give the **product rule**:

$$p(x, y) = p(x | y) p(y) = p(y | x) p(x).$$

This extends to higher dimensions, e.g.

$$p(w, x, y, z) = p(w, x, y | z) p(z) = p(w, x | y, z) p(y | z) p(z) = p(w | x, y, z) p(x | y, z) p(y | z) p(z).$$

## 9 Bayes' Rule

**Key Rule: Bayes' Rule**

Starting from the two forms of the product rule,  $p(x, y) = p(x | y)p(y) = p(y | x)p(x)$ , eliminating  $p(x, y)$  and rearranging gives:

$$p(y | x) = \frac{p(x | y) p(y)}{p(x)} = \frac{p(x | y) p(y)}{\int p(x | y) p(y) dy}$$

| Term       | Meaning                                                                                            |
|------------|----------------------------------------------------------------------------------------------------|
| $p(y)$     | <b>Prior</b> — what we know about $Y$ before seeing $X$ .                                          |
| $p(x   y)$ | <b>Likelihood</b> — propensity for observing a certain value of $X$ given a certain value of $Y$ . |
| $p(y   x)$ | <b>Posterior</b> — what we know about $Y$ after observing $X$ .                                    |
| $p(x)$     | <b>Evidence</b> — a constant that ensures the left-hand side is a valid distribution.              |

### Worked Example: The Apple/Orange Boxes (Bishop)

Let random variables  $B$  (box color) and  $F$  (fruit type) have  $Val(B) = \{r, b\}$  and  $Val(F) = \{a, o\}$ . **Red box:** 2 apples, 6 oranges. **Blue box:** 3 apples, 1 orange.

**Given:**

$$p(B=r) = 0.4, \quad p(B=b) = 0.6,$$

$$p(F=a | B=r) = 0.25, \quad p(F=o | B=r) = 0.75, \quad p(F=a | B=b) = 0.75, \quad p(F=o | B=b) = 0.25.$$

(a) **Overall probability of choosing an apple** (sum + product rule):

$$\begin{aligned} p(F=a) &= \sum_B p(F=a, B) = \sum_B p(F=a | B) p(B) \\ &= p(F=a | B=r)p(B=r) + p(F=a | B=b)p(B=b) \\ &= (0.25)(0.4) + (0.75)(0.6) = \mathbf{0.55}. \end{aligned}$$

(b) **Which box, given an orange was selected?** (Bayes' theorem):

$$\begin{aligned} p(B=r | F=o) &= \frac{p(F=o | B=r) p(B=r)}{p(F=o)} = \frac{p(F=o | B=r) p(B=r)}{1 - p(F=a)} = \frac{(0.75)(0.4)}{1 - 0.55} = \mathbf{0.667}, \\ p(B=b | F=o) &= 1 - p(B=r | F=o) = \mathbf{0.333}. \end{aligned}$$

**Conclusion:** The orange is more likely to have come from the **red box**!

## 10 Independence

### Definition : Independence

The independence of  $X$  and  $Y$  means that **every conditional distribution is the same** — the value of  $Y$  tells us nothing about  $X$ , and vice versa:

$$p(x | y) = p(x), \quad p(y | x) = p(y).$$

### Key Rule: Independence Factorization

When variables are independent, the joint distribution **factorizes** into a product of the marginals:

$$p(x, y) = p(x | y) p(y) = p(x) p(y).$$

## 11 Expectation

**Definition : Expectation**

The **expected (average) value** of some function  $f[x]$ , considering the distribution of  $X$ :

$$E[f[x]] = \sum_x f[x] p(x) \quad (\text{discrete}), \quad E[f[x]] = \int f[x] p(x) dx \quad (\text{continuous}).$$

**Key Rule: Rules of Expectation**

1. **Constant:**  $E[\kappa] = \kappa$ .
2. **Constant  $\times$  function:**  $E[\kappa f[x]] = \kappa E[f[x]]$ .
3. **Sum of functions:**  $E[f[x] + g[x]] = E[f[x]] + E[g[x]]$ .
4. **Product of functions (independent  $X, Y$ ):**  $E[f[x]g[y]] = E[f[x]] E[g[y]]$ , if  $X$  and  $Y$  are independent.

## 12 Common Probability Distributions

The choice of distribution depends on the **type/domain of data** to be modeled — but where do the probability *values* themselves come from? We use probability distributions defined over **parameters learned from data** (e.g. fitting a Normal distribution to population heights:  $\mu = 64$ ,  $\sigma^2 = 4$ ).

| Data Type                                                | Domain                                                                | Distribution                |
|----------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------|
| Univariate, discrete, binary                             | $x \in \{0, 1\}$                                                      | Bernoulli                   |
| Univariate, discrete, multi-valued                       | $x \in \{1, \dots, K\}$                                               | Categorical                 |
| Univariate, continuous, unbounded                        | $x \in \mathbb{R}$                                                    | Univariate Normal           |
| Univariate, continuous, bounded                          | $x \in [0, 1]$                                                        | Beta                        |
| Multivariate, continuous, unbounded                      | $\mathbf{x} \in \mathbb{R}^K$                                         | Multivariate Normal         |
| Multivariate, continuous, sums to one                    | $\sum_k x_k = 1$                                                      | Dirichlet                   |
| Bivariate ( $x_1 \in \mathbb{R}, x_2 \in \mathbb{R}^+$ ) | mixed                                                                 | Normal-scaled    Inv. Gamma |
| Vector $\mathbf{x}$ & PD matrix $\mathbf{X}$             | $\mathbf{x} \in \mathbb{R}^K, \mathbf{X} \in \mathbb{R}^{K \times K}$ | Normal    Inverse Wishart   |

**Bernoulli Distribution — single binary variable, parameter  $\lambda \in [0, 1]$** 

$$p(X=0 \mid \lambda) = 1 - \lambda, \quad p(X=1 \mid \lambda) = \lambda, \quad \text{or} \quad p(x) = \lambda^x (1 - \lambda)^{1-x} = \text{Bern}_x[\lambda].$$

*Example:* coin flip,  $X=1$  is ‘heads’,  $X=0$  is ‘tails’.

**Categorical Distribution** — 1-of- $K$  mutually exclusive states,  $K$  parameters  $\lambda_k \geq 0$ ,  $\sum_k \lambda_k = 1$

$x$  is a  $K$ -dimensional one-hot vector  $\mathbf{e}_k$  (e.g.  $K = 5$ ,  $x = \mathbf{e}_3 = [0, 0, 1, 0, 0]^\top$ ):

$$p(X=\mathbf{e}_k \mid \lambda) = \lambda_k \quad \text{or} \quad p(\mathbf{x}) = \prod_{k=1}^K \lambda_k^{x_k} = \lambda_k = \text{Cat}_x[\lambda].$$

**Univariate Normal (Gaussian)** — single continuous variable, parameters  $\mu \in \mathbb{R}$  (mean),  $\sigma^2 > 0$  (variance)

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) = \text{Norm}_x[\mu, \sigma^2].$$

**Multivariate Normal** —  $D$ -dim. variable, mean  $\mu \in \mathbb{R}^D$ , PD covariance  $\Sigma \in \mathbb{R}_+^{D \times D}$

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{D/2} |\Sigma|^{1/2}} \exp\left\{-0.5(\mathbf{x} - \mu)^\top \Sigma^{-1}(\mathbf{x} - \mu)\right\} = \text{Norm}_x[\mu, \Sigma].$$

**Types of covariance:** *spherical* ( $\Sigma = \sigma^2 I$ ), *diagonal* ( $\Sigma = \text{diag}(\sigma_1^2, \sigma_2^2)$ ), and *full* (general symmetric PD matrix) — controlling the shape/orientation of the Gaussian's contours.

## 13 Conjugate Distributions

### Definition : Conjugate Distribution

A **conjugate distribution** models the *parameters* of a probability distribution. The product of a probability distribution and its conjugate has the **same functional form as the conjugate**, times a constant. The parameters of conjugate distributions are called **hyperparameters**, since they control the parameter distributions.

| Distribution        | Domain                        | Parameters modeled by  |
|---------------------|-------------------------------|------------------------|
| Bernoulli           | $x \in \{0, 1\}$              | Beta                   |
| Categorical         | $x \in \{1, \dots, K\}$       | Dirichlet              |
| Univariate Normal   | $x \in \mathbb{R}$            | Normal Inverse Gamma   |
| Multivariate Normal | $\mathbf{x} \in \mathbb{R}^K$ | Normal Inverse Wishart |

### 13.1 Why Conjugacy Matters

- Learning parameters  $\theta$ :** recall Bayes' rule  $p(\theta \mid x) = \frac{p(x \mid \theta)p(\theta)}{\int p(x \mid \theta)p(\theta) d\theta}$ . If we *choose a prior*  $p(\theta)$  *conjugate to the likelihood*, the posterior must have the **same closed form** as the conjugate prior — which in turn implies the evidence integral collapses to a constant  $\kappa$ .
- Marginalizing over parameters (prediction):**  $p(x^* \mid \mathbf{x}) = \int p(x^* \mid \theta) p(\theta \mid \mathbf{x}) d\theta$ . If  $p(\theta \mid \mathbf{x})$  is chosen conjugate to  $p(x^* \mid \theta)$ , the product inside the integral becomes a **constant times a distribution**, and

$$\int \kappa \cdot p(\theta) d\theta = \kappa \int p(\theta) d\theta = \kappa \times 1 = \kappa,$$

making the predictive distribution tractable in closed form.

**Derivation Sketch**

$$\begin{aligned}
p(x^* | x) &= \frac{p(x^*, x)}{p(x)} && \text{(conditional probability)} \\
&= \frac{\int p(x^*, x, \theta) d\theta}{p(x)} && \text{(marginal probability)} \\
&= \frac{\int p(x^*, \theta | x) p(x) d\theta}{p(x)} && \text{(conditional probability)} \\
&= \int p(x^* | x, \theta) p(\theta | x) d\theta && \text{(conditional probability)} \\
&= \int p(x^* | \theta) p(\theta | x) d\theta && \text{(conditional independence)}
\end{aligned}$$

**13.2 Beta Distribution — conjugate of Bernoulli****Beta( $\lambda | \alpha, \beta$ ), hyperparameters  $\alpha, \beta > 0$** 

$$p(\lambda) = \frac{\Gamma[\alpha + \beta]}{\Gamma[\alpha]\Gamma[\beta]} \lambda^{\alpha-1} (1 - \lambda)^{\beta-1} = \text{Beta}_\lambda[\alpha, \beta], \quad \Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt, \quad \Gamma(n) = (n-1)!$$

**13.3 Dirichlet Distribution — conjugate of Categorical****Dir( $\lambda_1, \dots, \lambda_K | \alpha_1, \dots, \alpha_K$ ),  $K$  hyperparameters  $\alpha_k > 0$** 

$$p(\lambda_1, \dots, \lambda_K) = \frac{\Gamma\left[\sum_{k=1}^K \alpha_k\right]}{\prod_{k=1}^K \Gamma[\alpha_k]} \prod_{k=1}^K \lambda_k^{\alpha_k-1} = \text{Dir}_{\lambda_{1\dots K}}[\alpha_1, \dots, \alpha_K]$$

**13.4 Normal Inverse Gamma — conjugate of Univariate Normal****NormInvGam( $\mu, \sigma^2 | \alpha, \beta, \gamma, \delta$ ), four hyperparameters  $\alpha, \beta, \gamma > 0, \delta \in \mathbb{R}$** 

$$p(\mu, \sigma^2) = \frac{\sqrt{\gamma}}{\sigma \sqrt{2\pi}} \frac{\beta^\alpha}{\Gamma[\alpha]} \left(\frac{1}{\sigma^2}\right)^{\alpha+1} \exp\left[-\frac{2\beta + \gamma(\delta - \mu)^2}{2\sigma^2}\right] = \text{NormInvGam}_{\mu, \sigma^2}[\alpha, \beta, \gamma, \delta]$$

**13.5 Normal Inverse Wishart — conjugate of Multivariate Normal****NorIWis( $\mu, \Sigma | \alpha, \Psi, \gamma, \delta$ ), hyperparameters: scalar  $\alpha > 0$ , PD matrix  $\Psi \in \mathbb{R}_+^{D \times D}$ , scalar  $\gamma > 0$ , vector  $\delta \in \mathbb{R}^D$** 

$$p(\mu, \Sigma) = \frac{\gamma^{D/2} |\Psi|^{\alpha/2} \exp\left[-0.5 \left(\text{Tr}[\Psi \Sigma^{-1}] + \gamma(\mu - \delta)^\top \Sigma^{-1} (\mu - \delta)\right)\right]}{2^{\alpha D/2} (2\pi)^{D/2} |\Sigma|^{(\alpha+D+2)/2} \Gamma_D[\alpha/2]}$$

where the multivariate gamma function is  $\Gamma_D[a] = \pi^{a(a-1)/4} \prod_{j=1}^a \Gamma[a + (1-j)/2]$ . Intuitively:  $\alpha$  controls dispersion of covariances,  $\Psi$  the average covariance,  $\gamma$  the dispersion of mean vectors, and  $\delta$  the average mean vector.

**14 Worked Examples: Conjugate Updating for the Gaussian**

**Worked Example: 1 — Posterior of  $(\mu, \sigma^2)$  for a Univariate Gaussian**

**Find:** the posterior distribution of the parameters  $(\mu, \sigma)$  of a univariate Gaussian, given i.i.d. observations  $x[1], \dots, x[N]$ .

**Solution:** With a NormInvGam $(\alpha, \beta, \gamma, \delta)$  prior on  $\theta = (\mu, \sigma^2)$ ,

$$p(\theta | x) = \frac{\prod_{i=1}^N p(x[i] | \theta) p(\theta)}{p(x)} = \frac{\prod_{i=1}^N \text{Norm}_{x[i]}[\mu, \sigma^2] \text{NormInvGam}_{\mu, \sigma^2}[\alpha, \beta, \gamma, \delta]}{\int \prod_{i=1}^N \text{Norm}_{x[i]}[\mu, \sigma^2] \text{NormInvGam}_{\mu, \sigma^2}[\alpha, \beta, \gamma, \delta] d\theta}.$$

Rearranging the numerator into **constant**  $\times$  NormInvGam form:

$$\prod_{i=1}^N p(x[i] | \theta) p(\theta) = \kappa[\alpha, \beta, \gamma, \delta, x] \text{NormInvGam}_{\mu, \sigma^2}[\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}]$$

with updated hyperparameters

$$\tilde{\alpha} = \alpha + \frac{N}{2}, \quad \tilde{\gamma} = \gamma + N, \quad \tilde{\delta} = \frac{\gamma\delta + \sum_i x[i]}{\gamma + N}, \quad \tilde{\beta} = \frac{\sum_i x[i]^2}{2} + \beta + \frac{\gamma\delta^2}{2} - \frac{(\gamma\delta + \sum_i x[i])^2}{2(\gamma + N)}.$$

Substituting back into Bayes' rule, the constant  $\kappa$  cancels with the normalizing integral (which evaluates to 1), leaving:

$$p(\theta | x) = \text{NormInvGam}_{\mu, \sigma^2}[\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}]$$

i.e. the posterior has **exactly the same form** as the conjugate prior — Normal Inverse Gamma!

**Worked Example: 2 — Predictive Distribution  $p(x^* | x)$  for a Univariate Gaussian**

**Find:**  $p(x^* | \mathbf{x})$ , the distribution of a new data point given the observed data.

**Solution:** Marginalizing over the (posterior) parameter distribution from Example 1:

$$\begin{aligned} p(x^* | x) &= \iint p(x^* | \mu, \sigma^2) p(\mu, \sigma^2 | x) d\mu d\sigma^2 \\ &= \iint \text{Norm}_{x^*}[\mu, \sigma^2] \text{NormInvGam}_{\mu, \sigma^2}[\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}] d\mu d\sigma^2. \end{aligned}$$

Again writing the integrand as constant  $\times$  NormInvGam (which integrates to 1):

$$p(x^* | x) = \kappa[\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}, x^*] = \frac{1}{\sqrt{2\pi}} \frac{\tilde{\beta}^{\tilde{\alpha}} \sqrt{\tilde{\gamma}}}{\tilde{\beta}^{\tilde{\alpha}} \sqrt{\tilde{\gamma}}} \cdot \frac{\Gamma[\tilde{\alpha}]}{\Gamma[\tilde{\alpha}]},$$

where

$$\check{\alpha} = \tilde{\alpha} + \frac{1}{2}, \quad \check{\gamma} = \tilde{\gamma} + 1, \quad \check{\beta} = \frac{x^{*2}}{2} + \tilde{\beta} + \frac{\tilde{\gamma}\tilde{\delta}^2}{2} - \frac{(\tilde{\gamma}\tilde{\delta} + x^*)^2}{2(\tilde{\gamma} + 1)}.$$

This is a **constant** (a function of  $\{x^*, x[1], \dots, x[N]\}$  only) — the predictive distribution is tractable in closed form, thanks to conjugacy!

**15 Summary**

### Key Takeaways

1. **Probability spaces**  $(\Omega, E, P)$  formalize randomness; **random variables**  $X$  map outcomes to values, described by **joint probabilities**  $p(x, y)$ .
2. The core rules of probability are the **sum rule** (marginalization), **product rule** (conditioning), **Bayes' rule**, the notion of **independence**, and **expectation**.
3. Common distributions — **Bernoulli**, **categorical**, **univariate** and **multivariate normal** — are chosen based on the type/domain of the data.
4. **Conjugate distributions** (Beta, Dirichlet, Normal Inverse Gamma, Normal Inverse Wishart) model the *parameters* of these distributions and keep Bayesian posterior updates and predictive distributions in **closed form**.
5. These tools are the foundation for the graphical models (Bayesian networks, Markov random fields) and inference algorithms covered in the rest of the course.

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